Circles of Giving

Technical Summary

Final technical documentation for client
November 2026
https://circlesofgivingplatform.base44.app

1. Architecture Overview

Circles of Giving is a full-stack web application built on the Base44 platform-as-a-service. The frontend is a React single-page application; the backend consists of serverless functions, scheduled/event-driven workflows, and a document-based data layer with granular row-level security. The app is published as a responsive web app and wrapped for Android WebView deployment.

Request Flow: Browser -> React UI -> Base44 SDK -> (Entity CRUD / Backend Function / Integration) -> Response -> UI re-render (React Query cache + realtime subscriptions)

2. Technology Stack

Layer	Technology	Notes
Frontend	React 18 + Vite + Tailwind	shadcn/ui, lucide-react, framer-motion
Charts/Maps	Recharts + react-leaflet	Impact dashboards, OpenStreetMap
State/Data	TanStack Query + Base44 SDK	Cursor pagination, realtime subs
Forms/DnD	react-hook-form + pangea/dnd	Validated forms, drag-drop admin
Backend	Serverless functions + workflows	HTTP handlers, event triggers
Database	Base44 entities (document)	22 entities, row-level security
Auth	Base44 Auth + MFA	Email/password, Google, roles
Payments	Payments by Wix	Secure checkout, webhook verify
Integrations	Slack Bot + Supabase	OAuth-managed connectors
AI	In-app agents + InvokeLLM	Opportunity, registration, requests
Localization	Custom i18n EN/HE + RTL	Full dictionary, admin panel
Deployment	Base44 hosted + Android	Managed hosting, WebView mobile

3. Data Model (22 Entities)

The platform uses 22 entities spanning user management, community interaction, volunteer operations, and governance. Each entity is a JSON schema with built-in fields and custom properties.

User	**VolunteerProfile**	**Post**	**Comment**
Opportunity	**Event**	**WorkshopInquiry**	**ShabbatMeal**
HourLog	**Badge**	**Review**	**Group**
GroupMembership	**Message**	**SosRequest**	**HealthRequest**
EventMessage	**Notification**	**GalleryAlbum**	**GalleryPhoto**
CorporateInquiry	**RateLimit**		

4. Security Architecture

Security is enforced at multiple layers, from database access rules to payment verification:

- Field-level RLS - sensitive fields (phone, contact) restricted by visibility settings
- Payment verification - webhook JWT verification, idempotent grant, server-side price validation
- Moderation tools - post reporting, flagging, and admin review workflow
- Rate limiting entity to protect against abuse
- MFA support - optional two-factor authentication for app users
- OAuth integration security - connector tokens managed via platform

5. Backend Functions (21 Serverless Endpoints)

Function	Purpose
create-checkout	Initiates Wix checkout for donations
rsvpEvent	Event RSVP toggling and attendees
eventChat	Event-specific chat messages
enrollWorkshop	Workshop enrollment inquiries
togglePostLike	Toggle likes on feed posts
toggleFollow	Follow/unfollow profiles
reportPost	Flag posts for moderation
addOpportunityRemark	Add remarks to opportunities
helpRequestAction	Claim/resolve SOS & health
manageGroupMembership	Group join/leave/admin ops
manageHourLog	Submit/approve volunteer hours
generateCertificate	Create volunteer PDFs
notifyOpportunity	Notify on new opportunities
notifyMatchingOpportunities	Match profiles to opportunities
sendSlackNotification	Post updates to Slack
sendAdminAnnouncement	Distribute announcements
submitContactForm	Process contact form
promptMealReviews	Prompt Shabbat meal reviews
listPendingOpportunities	Pending ops for moderation
migrateCategories	Category schema migration
publicApi	Public API endpoint

6. Automated Workflows

Workflow	Trigger	Action
Notify Slack on New Event	Event created	Post to #social Slack
Notify Slack on New Opportunity	Opportunity created	Post to #social Slack
Notify Users on New Opportunity	Opportunity created	In-app notifications
Match Opportunities on Profile Update	Profile updated	Match + notify users
Prompt Meal Reviews on Completion	Meal status change	Review prompt to guests

7. AI Assistants

Agent	Role	Channel
Opportunity Matcher	Match volunteers to opportunities	In-app

Registration Assistant	Guide new member sign-up	In-app
Request Filing Assistant	File SOS and health requests	In-app

8. External Integrations

Slack (Slack Bot connector): Automated notifications to #social for new events, opportunities, and community activity. OAuth-managed with chat:write and channel:read scopes.

Payments by Wix: Secure donation checkout with webhook-verified payment confirmation, idempotent access grants, and server-side price validation. Checkout is public; access grants require verified PAID status.

Supabase (connector): Read-only database access for reporting and external integration.

App MCP Server: Exposes app data and backend functions to AI clients (ChatGPT, Claude) via MCP for external automation.

9. Localization & Accessibility

- Accessibility widget: high-contrast, link highlighting, readable spacing, pause animations
- Mobile-first responsive design optimized for Android WebView
- Overscroll behavior and text-selection controls for native mobile feel

10. Deployment & Infrastructure

- Android WebView wrapper for mobile distribution
- Secrets managed via platform secret store (payment keys, connector tokens)
- Realtime entity subscriptions for live updates across connected clients